

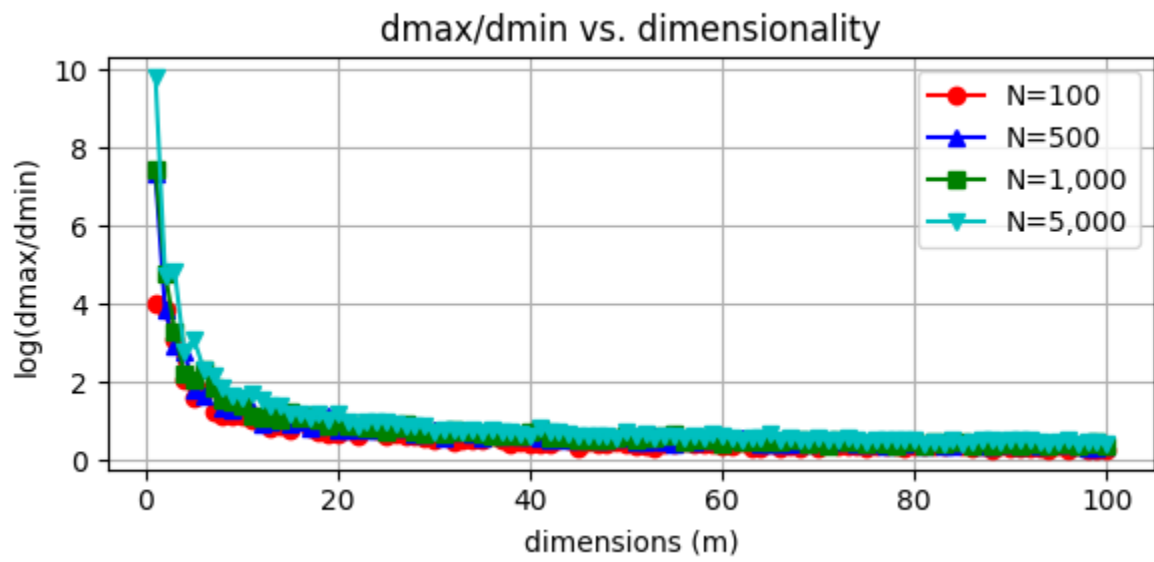
“Curse of dimensionality” is usually used to describe why learning in high dimensions is hard: as the number of features grows, the volume of the space explodes, data becomes sparse, and distance based methods like KNN lose contrast between near and far points. Our in part b we can see this happen. In high dimensions the ratio between the distance to the nearest and farthest neighbors collapse, so every point starts to look almost equally far from the query. At first glance this seems to show that high dimensionality is simply bad. However, recent work argues that the same geometric effects can also be a “blessing of dimensionality” when used in the right way.

One important reason is concentration of measure. For many high dimensional distributions, random points concentrate in a thin shell and most pairwise distances are close to their mean. This makes the geometry of the data more regular and predictable. Gorbunov and Tyukin show that in such settings, large sets of random points are, with high probability, linearly separable. Each point can be separated from all others by a simple linear functional such as a Fisher discriminant. In other words, if data really fill out a high-dimensional space in a reasonably “nice” way then adding dimensions can actually make it easier for linear classifiers to separate classes, not harder. High dimension becomes a resource that allows many simple separating hyperplanes to exist.

A second reason high dimension can help is when extra features carry real signal, not noise. Domingos emphasizes that the “curse” shows up when we imagine the data as being uniformly distributed over all possible combinations of attributes which is rarely true in practice. Real data are highly structured and clustered on low dimensional manifolds inside the larger feature space. He calls this the “blessing of non-uniformity”. Even though the ambient dimension is large, examples occupy only a tiny subset of that space so learning can still succeed as long as models and regularization exploit this structure. Modern high dimensional methods such as sparse regression, kernel machines, and deep networks implicitly rely on this idea. They search for simple structure (sparse weights, low-dimensional manifolds, or hierarchical features) inside a large ambient space rather than trying to fill the whole space with data.

Finally, several authors argue that curse and blessing are two sides of the same underlying phenomenon. The same concentration that makes distances similar, which is bad for naive nearest neighbor search, also implies that random vectors are almost orthogonal, which is good for constructing robust ensembles, averaging models, or using random projections. High dimensional limit theorems can therefore help explain why techniques like wide neural networks or random feature maps often generalize well despite operating in huge feature spaces. Rather than thinking of dimension as automatically harmful, it is more accurate to say that high dimensionality amplifies whatever structure or lack of structure the data already have. Thus, high dimensionality is neither a blessing nor a curse a priori. It becomes a curse if we rely on uniform coverage of the space and on simple distance heuristics oblivious to structure, as in our KNN experiment. It becomes a blessing when we exploit concentration, sparsity, and nonuniformity to make simple linear or low complexity models work effectively in very large feature spaces.

Plot:



Reference

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